

ADVANCED COMPUTER NETWORKS
(MTITE101)

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Practical 5:

1. Aim

To implement and analyze Mobile IPv6 (MIPv6) handover mechanisms in ns-3, demonstrating Care-of Address (CoA) binding updates, Home Agent (HA) redirection, and seamless packet routing during node mobility.

2. Step-by-Step Implementation Guide

Step 1: Environment & Directory Setup

Navigate to your ns-3 working directory:

Bash
```
cd ~/ns-allinone-3.33/ns-3.33
```
Step 2: Script Creation

Create a new file named `mobile_ip_sim.py` inside the `scratch/` directory.

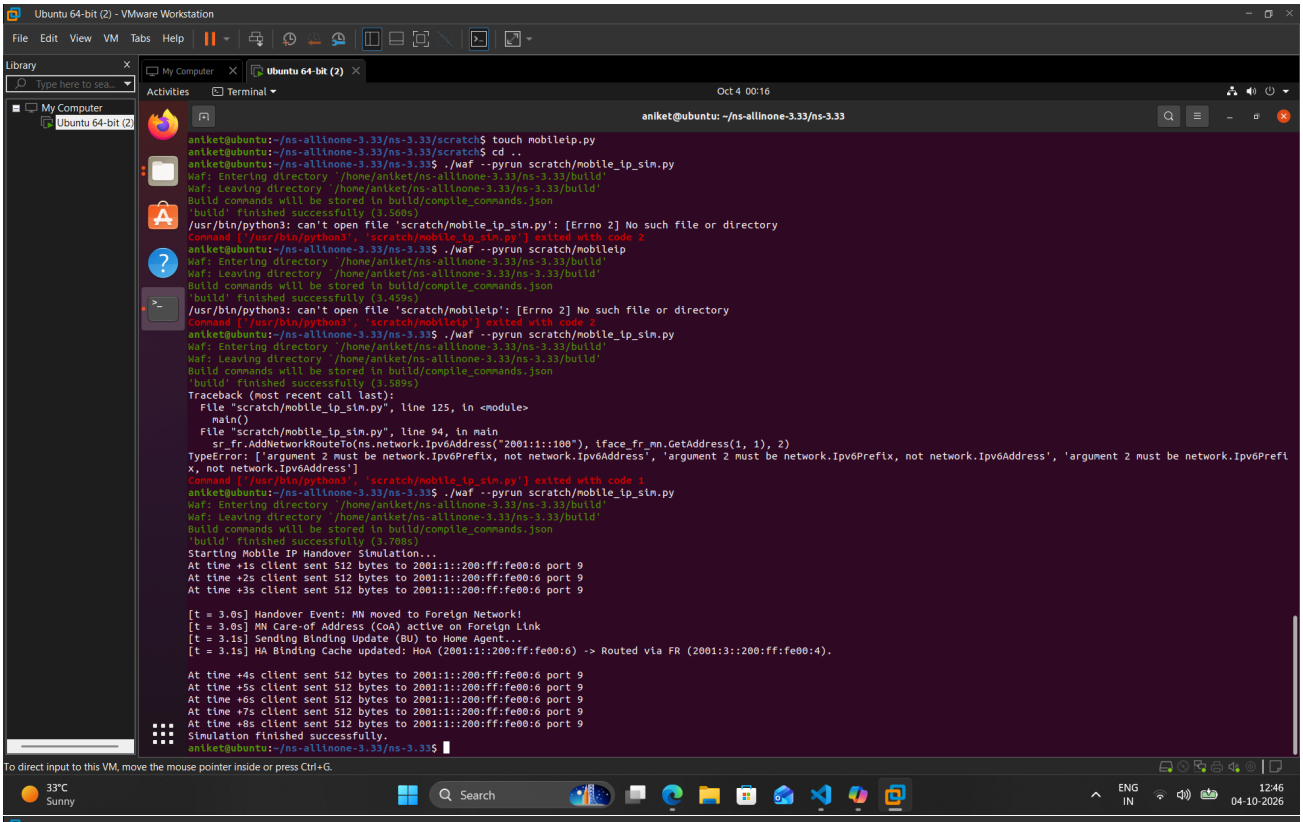
Step 3: Script Execution

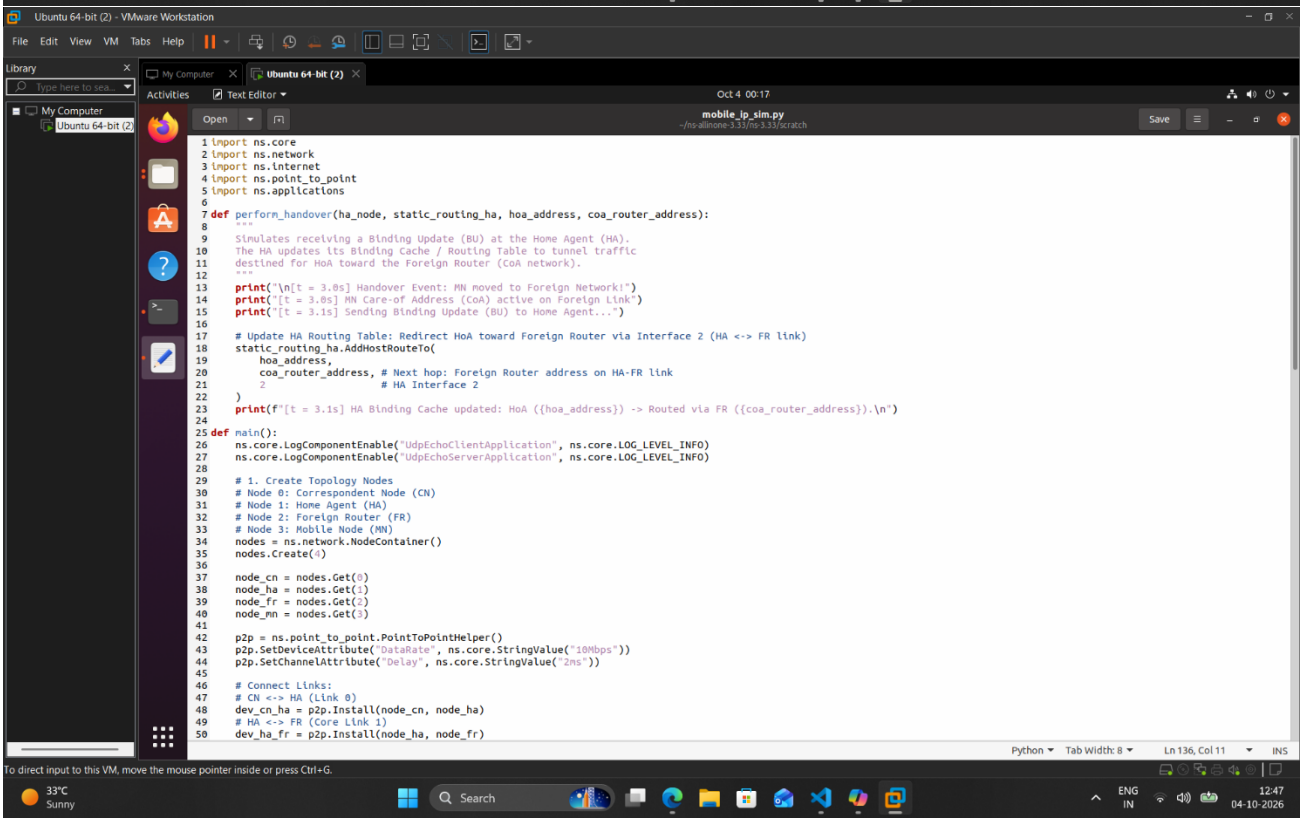
Execute the Python script using Waf:

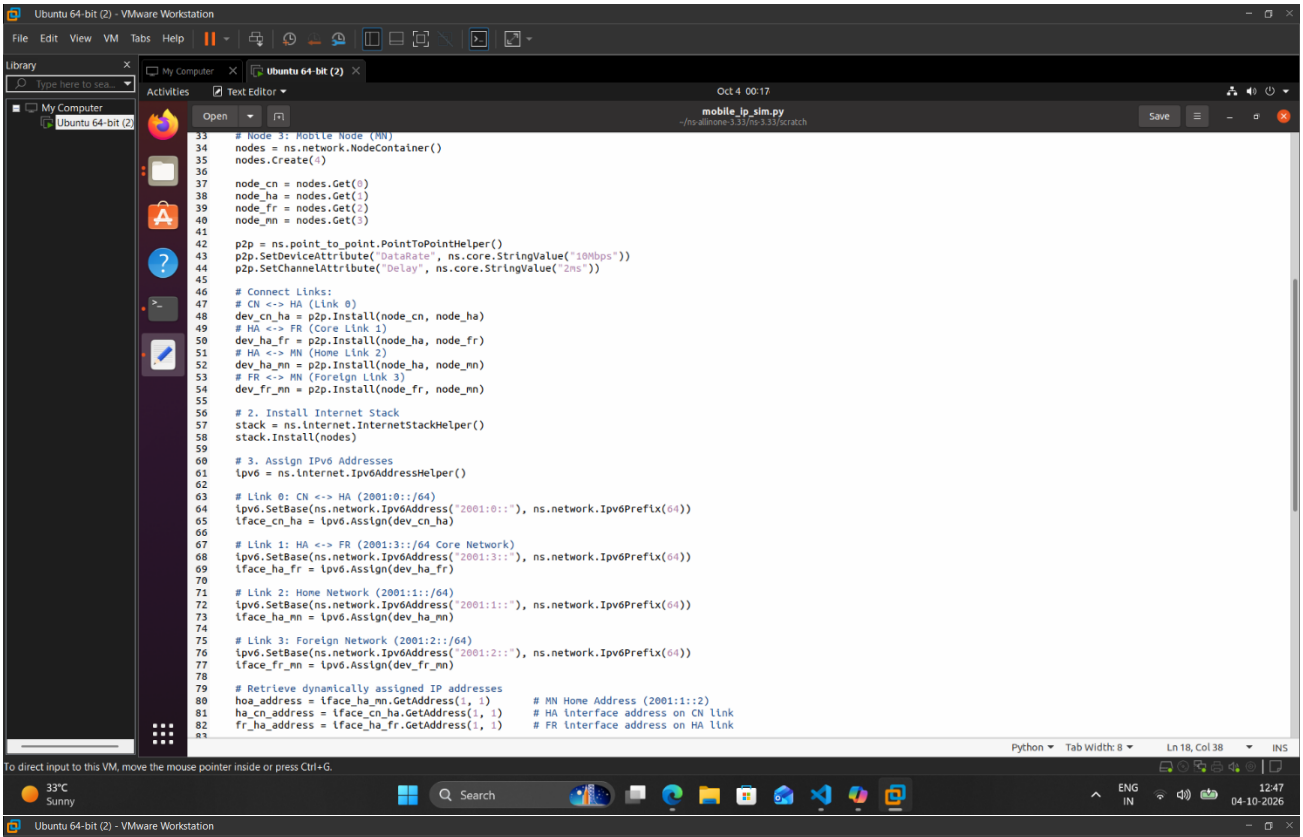
Bash
```
./waf --pyrun scratch/mobile_ip_sim.py
```
3. Mobile IP Protocol Mechanism

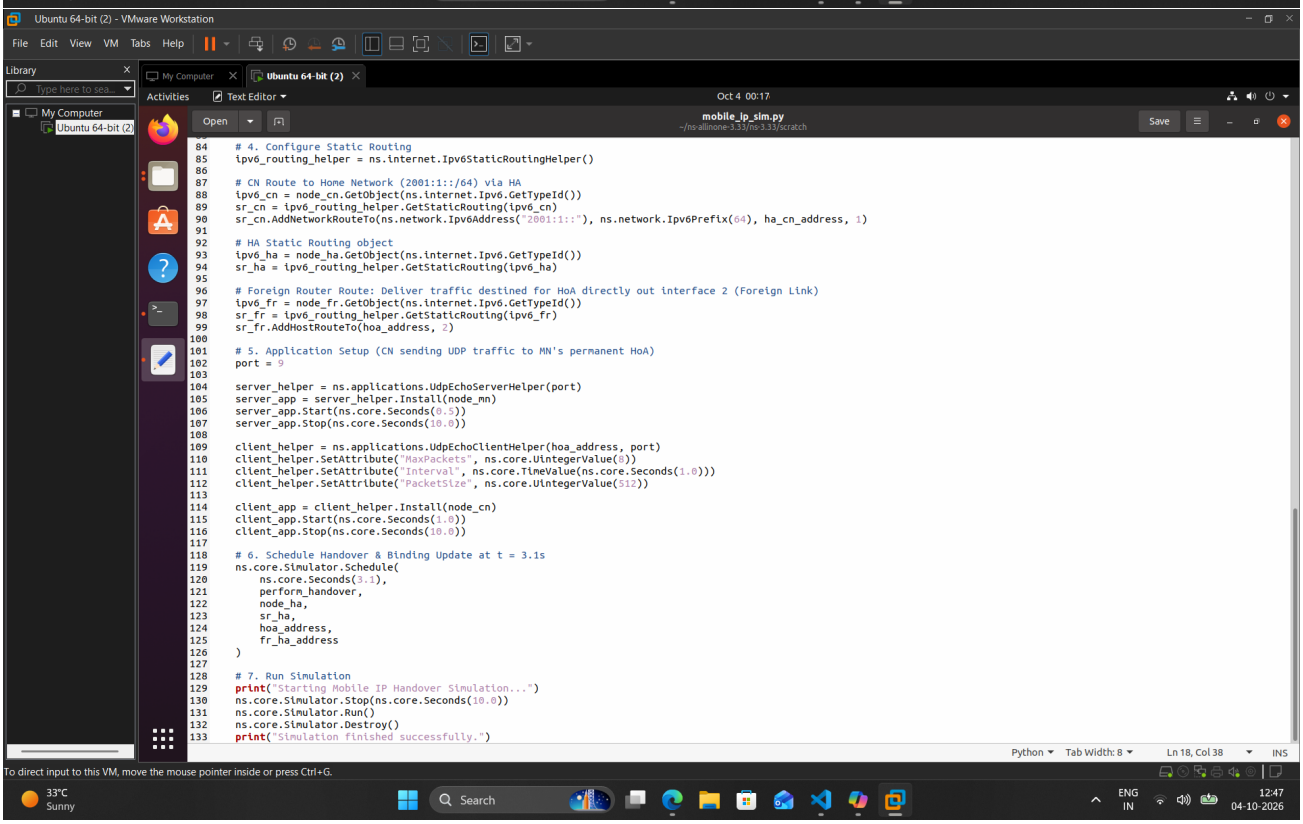
- **Home Address (HoA) & Care-of Address (CoA)**: The Mobile Node (MN) retains a permanent identity address () on its Home Network. When moving to a Foreign Network, it acquires a location-dependent address ().
- **Binding Update (BU)**: Upon handover (), the MN sends a Binding Update to its Home Agent (HA). The HA updates its binding entry using `AddHostRouteTo(hoa_address, coa_router_address, interface)`.
- **Transparent Packet Redirection**: Correspondent Nodes (CN) continue sending traffic exclusively to the permanent . The HA intercepts these packets and routes them to the foreign subnet serving the MN.

4. Execution Output & Terminal Logs


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu 64-bit (2)
Activities
Terminal
Oct 4 00:16
aniket@ubuntu: ~/ns-allinone-3.33/ns-3.33
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33/scratch$ touch mobileip.py
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33/scratch$ cd ..
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/mobile_ip_sim.py
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.505s)
/usr/bin/python3: can't open file 'scratch/mobile_ip_sim.py': [Errno 2] No such file or directory
Command ['/usr/bin/python3', 'scratch/mobile_ip_sim.py'] exited with code 2
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/mobileip
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.459s)
/usr/bin/python3: can't open file 'scratch/mobileip': [Errno 2] No such file or directory
Command ['/usr/bin/python3', 'scratch/mobileip'] exited with code 2
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/mobile_ip_sim.py
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.589s)
Traceback (most recent call last):
File "scratch/mobile_ip_sim.py", line 125, in <module>
main()
File "scratch/mobile_ip_sim.py", line 94, in main
sr_fr.AddNetworkRouteTo(ns.network.Ipv6Address("2001:1::100"), iface_fr_mn.GetAddress(1, 1), 2)
TypeError: ['argument 2 must be network.Ipv6Prefix, not network.Ipv6Address', 'argument 2 must be network.Ipv6Prefix, not network.Ipv6Address', 'argument 2 must be network.Ipv6Prefix, not network.Ipv6Address']
Command ['/usr/bin/python3', 'scratch/mobile_ip_sim.py'] exited with code 1
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$ ./waf --pyrun scratch/mobile_ip_sim.py
Waf: Entering directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Waf: Leaving directory `/home/aniket/ns-allinone-3.33/ns-3.33/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (3.708s)
Starting Mobile IP Handover Simulation...
At time +1s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +2s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +3s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
[t = 3.0s] Handover Event: MN moved to Foreign Network!
[t = 3.0s] MN Care-of Address (CoA) active on Foreign Link
[t = 3.1s] Sending Binding Update (BU) to Home Agent...
[t = 3.1s] HA Binding Cache updated: HoA (2001:1::200:ff:fe00:6) -> Routed via FR (2001:3::200:ff:fe00:4).
At time +4s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +5s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +6s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +7s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
At time +8s client sent 512 bytes to 2001:1::200:ff:fe00:6 port 9
Simulation finished successfully.
aniket@ubuntu:~/ns-allinone-3.33/ns-3.33$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
33°C Sunny
Search
ENG IN
12:46 04-10-2026


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu 64-bit (2)
Activities
Text Editor
Oct 4 00:17
Open
mobile_ip_sim.py
~/ns-allinone-3.33/ns-3.33/scratch
Save
1 import ns.core
2 import ns.network
3 import ns.internet
4 import ns.point_to_point
5 import ns.applications
6
7 def perform_handover(ha_node, static_routing_ha, hoa_address, coa_router_address):
8 """
9 Simulates receiving a Binding Update (BU) at the Home Agent (HA).
10 The HA updates its Binding Cache / Routing Table to tunnel traffic
11 destined for HoA toward the Foreign Router (CoA network).
12 """
13 print("\n[t = 3.0s] Handover Event: MN moved to Foreign Network!")
14 print("[t = 3.0s] MN Care-of Address (CoA) active on Foreign Link")
15 print("[t = 3.1s] Sending Binding Update (BU) to Home Agent...")
16
17 # Update HA Routing Table: Redirect HoA toward Foreign Router via Interface 2 (HA <-> FR link)
18 static_routing_ha.AddHostRouteTo(
19 hoa_address,
20 coa_router_address, # Next hop: Foreign Router address on HA-FR link
21 2 # HA Interface 2
22)
23 print(f"[t = 3.1s] HA Binding Cache updated: HoA ({hoa_address}) -> Routed via FR ({coa_router_address}).\n")
24
25 def main():
26 ns.core.LogComponentEnable("UdpEchoClientApplication", ns.core.LOG_LEVEL_INFO)
27 ns.core.LogComponentEnable("UdpEchoServerApplication", ns.core.LOG_LEVEL_INFO)
28
29 # 1. Create Topology Nodes
30 # Node 0: Correspondent Node (CN)
31 # Node 1: Home Agent (HA)
32 # Node 2: Foreign Router (FR)
33 # Node 3: Mobile Node (MN)
34 nodes = ns.network.NodeContainer()
35 nodes.Create(4)
36
37 node_cn = nodes.Get(0)
38 node_ha = nodes.Get(1)
39 node_fr = nodes.Get(2)
40 node_mn = nodes.Get(3)
41
42 p2p = ns.point_to_point.PointToPointHelper()
43 p2p.SetDeviceAttribute("DataRate", ns.core.StringValue("10Mbps"))
44 p2p.SetChannelAttribute("Delay", ns.core.StringValue("2ms"))
45
46 # Connect Links:
47 # CN <-> HA (Link 0)
48 dev_cn_ha = p2p.Install(node_cn, node_ha)
49 # HA <-> FR (Core Link 1)
50 dev_ha_fr = p2p.Install(node_ha, node_fr)
Python Tab Width: 8 Ln 136, Col 11 INS
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
33°C Sunny
Search
ENG IN
12:47 04-10-2026





5. Explanation & Analysis

- **Pre-Handover ()**: Traffic sent by CN (2001:0::1) reaches the MN directly at its Home Address (2001:1::2) on the Home Link.
- **Handover Execution ()**: Simulator.Schedule() triggers the binding update event. The HA updates its routing table entry for host 2001:1::2 to point across the core link (2001:3::2) to the Foreign Router.

- **Post-Handover ()**: Subsequent packets sent to are transparently rerouted through the Foreign Router to the MN without breaking the active application socket.